

Mobile Development

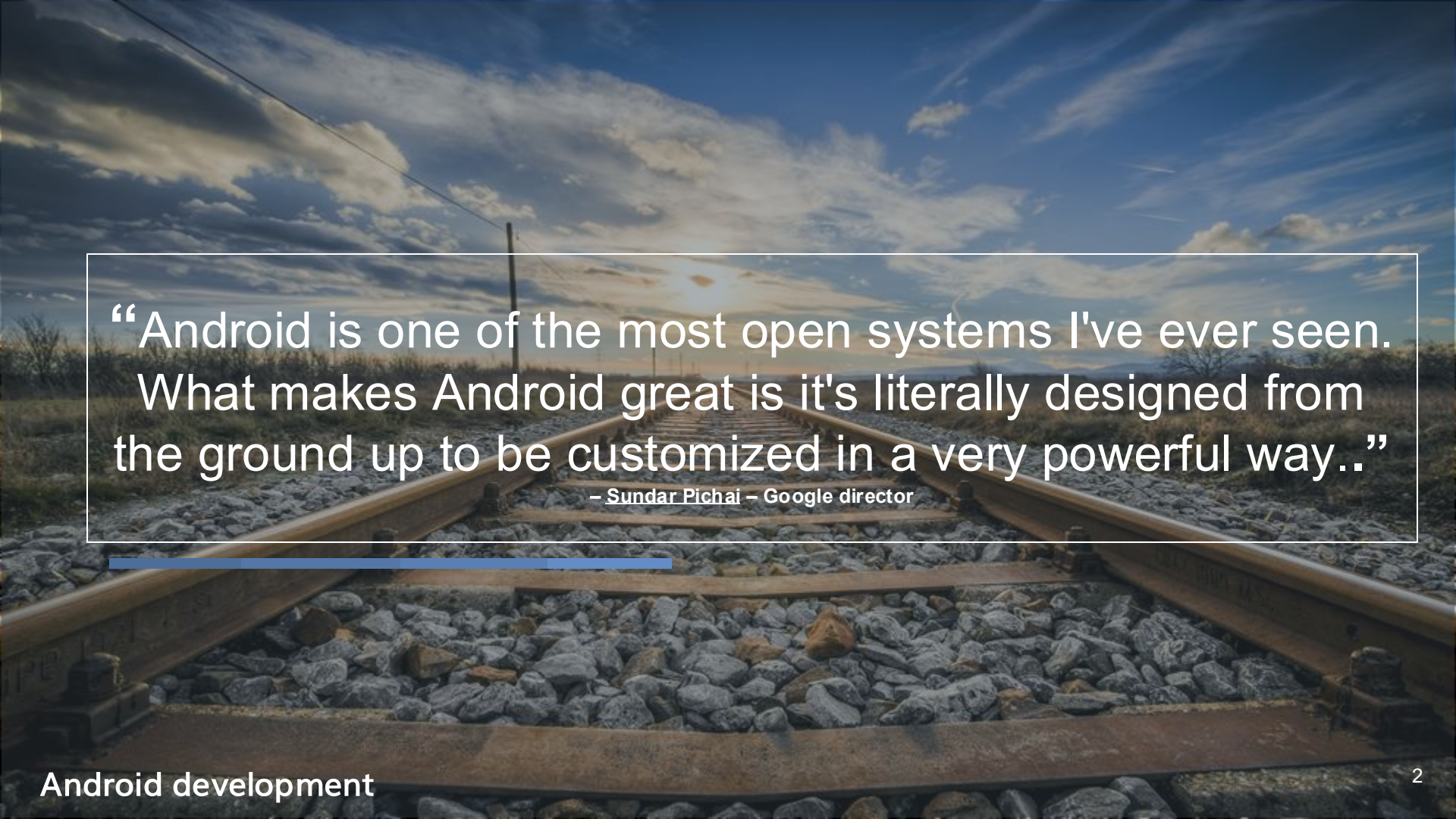
(Lab 4)

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Smart Software System Team



“Android is one of the most open systems I've ever seen. What makes Android great is it's literally designed from the ground up to be customized in a very powerful way..”

– Sundar Pichai – Google director

Learning Objectives



- Provide you with realistic, hands-on experience developing Android applications.
- Create a portfolio of apps that you can show your friends, discuss in interviews, and borrow for other applications.

Storage



Shared preferences

most commonly used for storing app preferences, as key-value pairs

Database

for storing structured data in a private database

File storage

for storing all kinds of media/documents to the file system

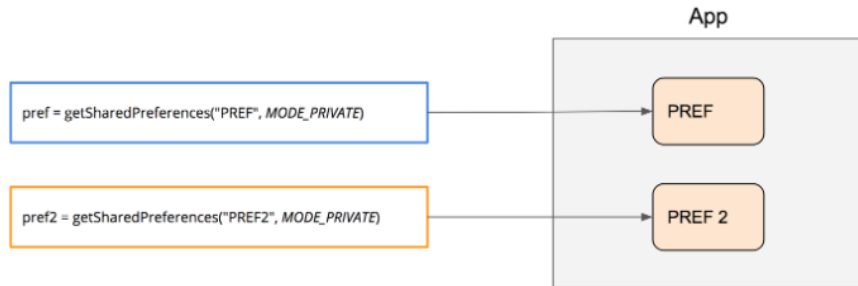
Shared Preferences



```
val pref = getSharedPreferences("PREF", MODE_PRIVATE)
```



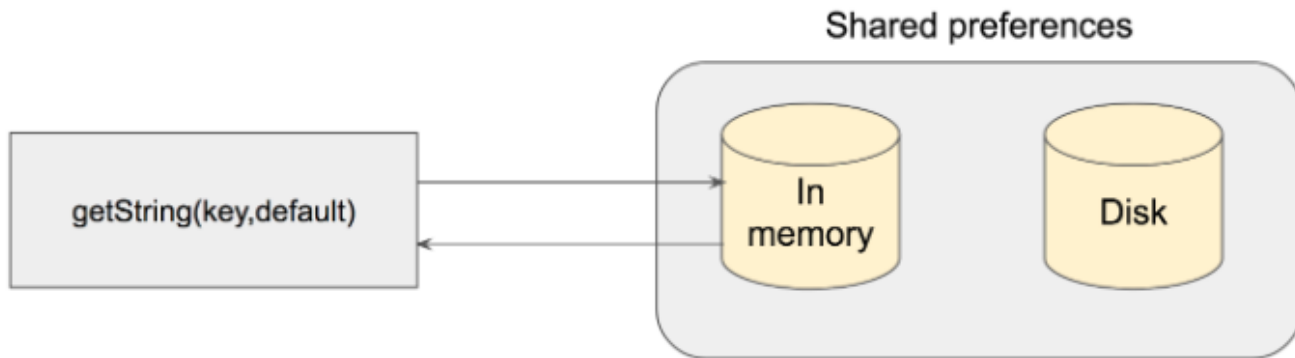
```
val pref2 = getSharedPreferences("PREF2", MODE_PRIVATE)
```



Shared Preferences – Get data



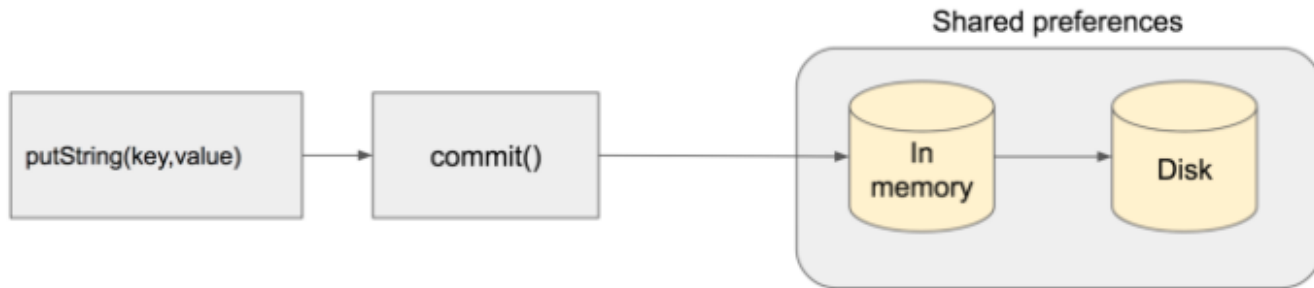
```
val value = pref.getString(key, null)
```



Shared Preferences – Save data via commit



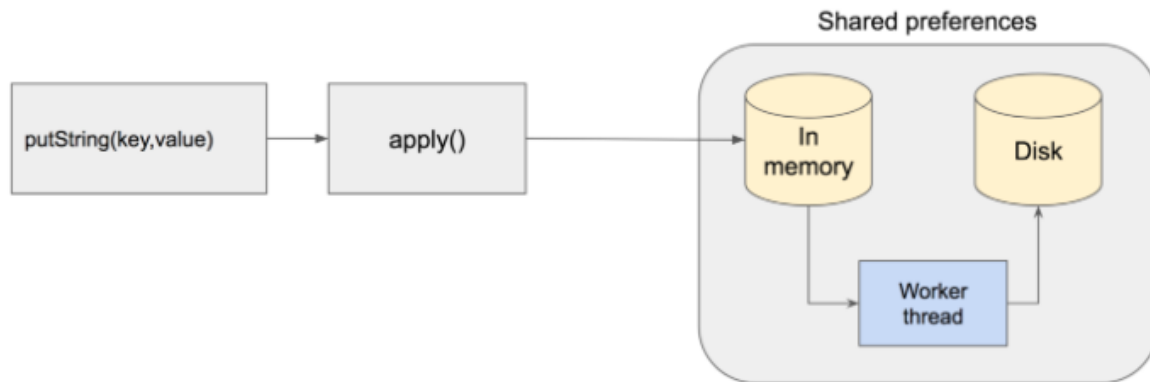
```
val result = pref.edit().putString(key, "value").commit()
```



Shared Preferences – Save data via apply



```
pref.edit().putString(key, "value").apply()
```





Shared Preferences – Read data

```
// Retrieving the value using its keys the file name  
// must be same in both saving and retrieving the data  
SharedPreferences sh = getSharedPreferences("MySharedPref", MODE_APPEND);  
  
// The value will be default as empty string because for  
// the very first time when the app is opened, there is nothing to show  
String s1 = sh.getString("name", "");  
int a = sh.getInt("age", 0);  
  
// We can then use the data  
name.setText(s1);  
age.setText(String.valueOf(a));
```



Shared Preferences – Write data

```
// Storing data into SharedPreferences  
SharedPreferences sharedPreferences = getSharedPreferences("MySharedPref",MODE_PRIVATE);  
  
// Creating an Editor object to edit(write to the file)  
SharedPreferences.Editor myEdit = sharedPreferences.edit();  
  
// Storing the key and its value as the data fetched from edittext  
myEdit.putString("name", name.getText().toString());  
myEdit.putInt("age", Integer.parseInt(age.getText().toString()));  
  
// Once the changes have been made,  
// we need to commit to apply those changes made,  
// otherwise, it will throw an error  
myEdit.commit();
```

Shared Preferences - Example



```
public class MainActivity extends AppCompatActivity {
    private EditText name, age;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        name = findViewById(R.id.edit1);
        age = findViewById(R.id.edit2);
    }
    @Override
    protected void onResume() {
        super.onResume();
        SharedPreferences sh = getSharedPreferences("MySharedPref", MODE_PRIVATE);
        String s1 = sh.getString("name", "");
        int a = sh.getInt("age", 0);
        name.setText(s1);
        age.setText(String.valueOf(a));
    }
    @Override
    protected void onPause() {
        super.onPause();
        SharedPreferences sharedPreferences = getSharedPreferences("MySharedPref", MODE_PRIVATE);
        SharedPreferences.Editor myEdit = sharedPreferences.edit();
        myEdit.putString("name", name.getText().toString());
        myEdit.putInt("age", Integer.parseInt(age.getText().toString()));
        myEdit.apply();
    }
}
```

Internal and external storage



Internal

Phone memory

External

SD or Micro SD Card



Internal storage – Write and read data

```
FileOutputStream fOut = openFileOutput("file name here",MODE_WORLD_READABLE);
```

```
String str = "data";  
fOut.write(str.getBytes()); fOut.close();
```

```
FileInputStream fin = openFileInput(file);
```

```
int c; String temp="";  
while( (c = fin.read()) != -1){  
    temp = temp + Character.toString((char)c);  
}  
fin.close();
```

```

public void onClick(View v) {
    String myData = "";
    switch (v.getId()) {
        case R.id.btnSave:
            try {
                FileOutputStream fos = new FileOutputStream(myInternalFile);
                fos.write(myInputText.getText().toString().getBytes());
                fos.close();
            } catch (IOException e) {
                e.printStackTrace();
            }
            myInputText.setText("");
            responseText
                .setText("Write data successfully");
            break;
        case R.id.btnDisplay:
            try {
                FileInputStream fis = new FileInputStream(myInternalFile);
                DataInputStream in = new DataInputStream(fis);
                BufferedReader br = new BufferedReader(
                    new InputStreamReader(in));
                String strLine;
                while ((strLine = br.readLine()) != null) {
                    myData = myData + strLine;
                }
                in.close();
            } catch (IOException e) {
                e.printStackTrace();
            }
            myInputText.setText(myData);
            responseText
                .setText("Get data successfully");
            break;
    }
}

```

Internal storage





External storage - Permission

```
<manifest ... >
    <uses-permission
android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
    <uses-permission
android:name="android.permission.READ_EXTERNAL_STORAGE"/>

    <application ... >
        <activity android:name=".MainActivity" ... >
            ...
        </activity>
    </application>
</manifest>
```

```
private int EXTERNAL_STORAGE_PERMISSION_CODE = 23;
```

```
EditText editText;
```

```
@Override
```

```
protected void onCreate(Bundle savedInstanceState) {
```

```
    super.onCreate(savedInstanceState);
```

```
    setContentView(R.layout.activity_main);
```

```
    editText = (EditText) findViewById(R.id.editText_data);
```

```
    File folder = Environment.getExternalStoragePublicDirectory(Environment.DIRECTORY_DOWNLOADS);
```

```
    File file = new File(folder, "UIT.txt");
```

```
    writeTextData(file, editTextData);
```

```
}
```

```
private void writeTextData(File file, String data) {
```

```
    FileOutputStream fileOutputStream = null;
```

```
    try {
```

```
        fileOutputStream = new FileOutputStream(file);
```

```
        fileOutputStream.write(data.getBytes());
```

```
        Toast.makeText(this, "Done" + file.getAbsolutePath(), Toast.LENGTH_SHORT).show();
```

```
    } catch (Exception e) {
```

```
        e.printStackTrace();
```

```
    } finally {
```

```
        if (fileOutputStream != null) {
```

```
            try {
```

```
                fileOutputStream.close();
```

```
            } catch (IOException e) {
```

```
                e.printStackTrace();
```

```
            }
```

```
        }
```

```
    }
```

External storage



Database



- **Oracle**
- **Microsoft**
 - **SQL Server** (powerful)
 - **Access** (simple)
- **PostgreSQL**
 - powerful/complex free open-source database system
- **SQLite**
 - transportable, lightweight free open-source database system
- **MySQL**
 - simple free open-source database system
 - many servers run "LAMP" (Linux, Apache, MySQL, and PHP)
 - Wikipedia is run on PHP and MySQL



Android SQLite Database



```
SQLiteDatabase db = openOrCreateDatabase(  
    "name", MODE_PRIVATE, null);  
db.execSQL("SQL query");
```

```
ContentValues cvalues = new ContentValues();  
cvalues.put("columnName1", value1);  
cvalues.put("columnName2", value2);  
...  
db.insert("tableName", null, cvalues);
```

```
db.execSQL("INSERT INTO tableName ("  
    + columnName1 + ", " + columnName2  
    + ") VALUES (" + value1 + ", " + value2 + ")");
```

Android SQLite Database – Cursor



Cursor lets you iterate through row results one at a time

— — —

```
Cursor cursor = db.rawQuery("SELECT * FROM students");
cursor.moveToFirst();
do {
    int id = cursor.getInt(cursor.getColumnIndex("id"));
    String email = cursor.getString(
        cursor.getColumnIndex("email"));
    ...
} while (cursor.moveToNext());
cursor.close();
```

Dictionary app exercise



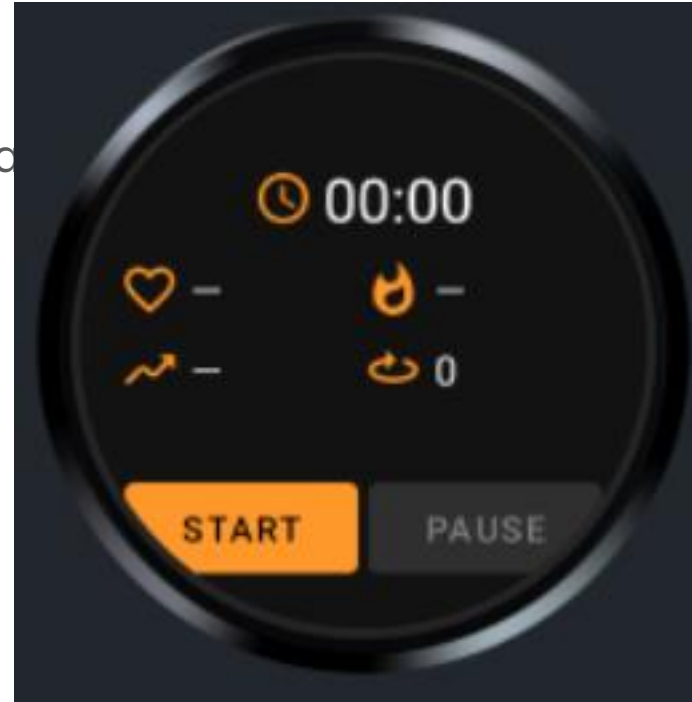
- Write an app that lets the user look up words in a dictionary.
 - The dictionary should be created as a SQLite **database**.
 - When the user types in a word, if that exact word exists in the dictionary, show its definition.
 - If the exact word does not exist in the dictionary, list all words of which the user's text is a substring.

Or use your assignment's app if it's already used SQLite database.



Exercise

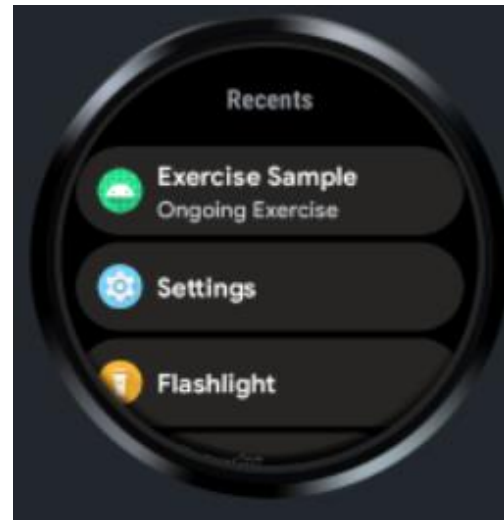
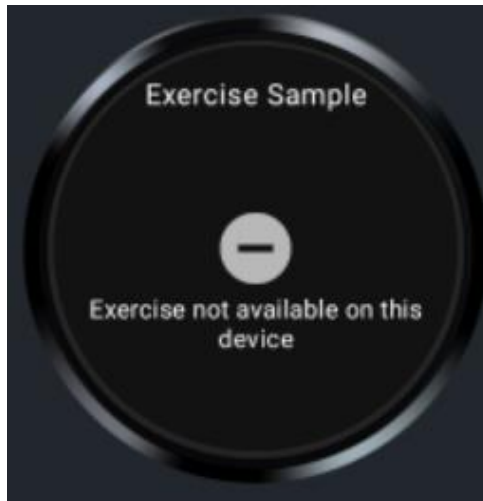
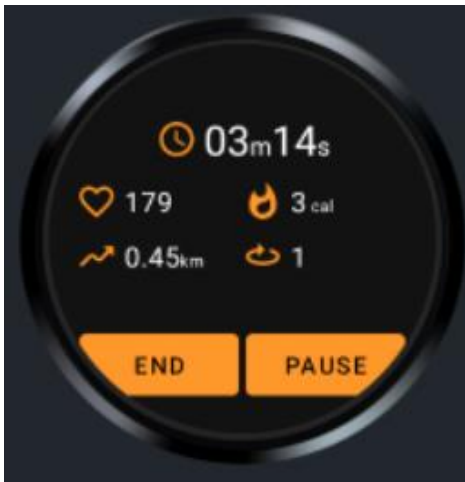
- Build your own Health measure application
- Example of this app is on the picture.



Exercise

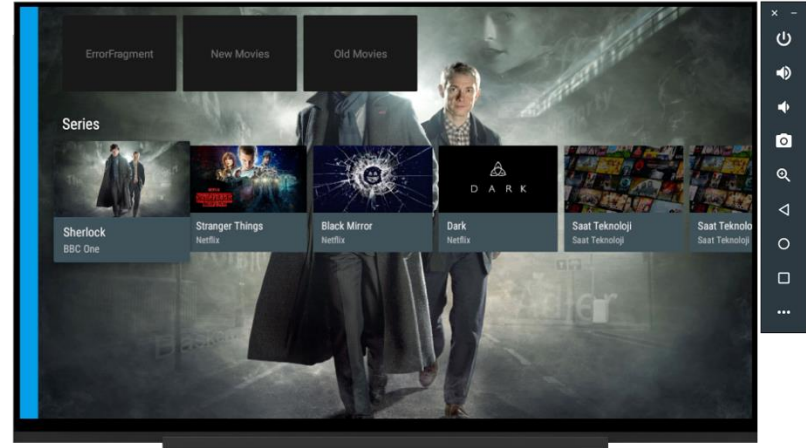


Hint: <https://developer.android.com/training/wearables/health-services>



Homework

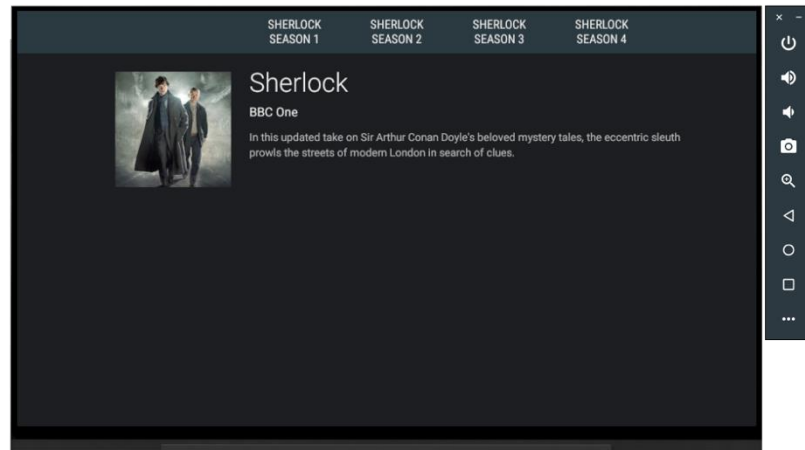
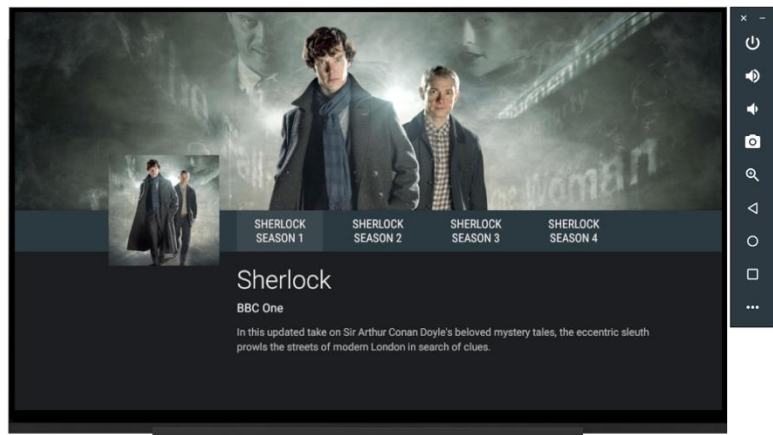
- Build your own TV Android application with your favorite movie.
- Example of this app is on the picture.
- You can use this API to load film:
<https://developers.themoviedb.org/3>



Homework



<https://developer.android.com/training/tv/start/start>



HW – Hint – Android Manifest



```
<uses-permission android:name="android.permission.INTERNET" />

<!-- TV app need to declare touchscreen not required -->
<uses-feature
    android:name="android.hardware.touchscreen"
    android:required="false" />

<!--
true:  your app runs on only TV
false: your app runs on phone and TV
-->
<uses-feature
    android:name="android.software.leanback"
    android:required="true" />
```

HW – Hint – Android Manifest



```
<uses-permission android:name="android.permission.INTERNET" />
<uses-permission android:name="android.permission.RECORD_AUDIO"/>
<uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />
```

```
<!-- TV app need to declare touchscreen not required -->
```

```
<uses-feature
```

```
    android:name="android.hardware.touchscreen"
```

```
    android:required="false" />
```

```
<!--
```

```
    true:  your app runs on only TV
```

```
    false: your app runs on phone and TV
```

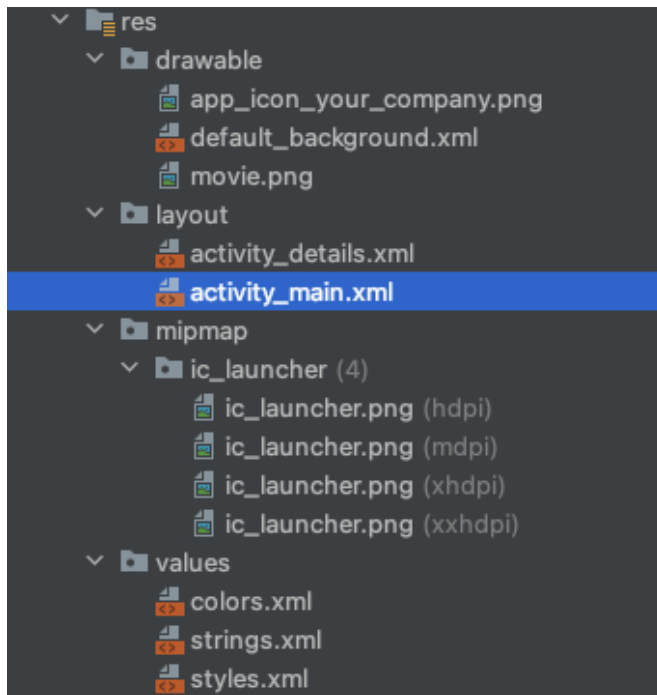
```
-->
```

```
<uses-feature
```

```
    android:name="android.software.leanback"
```

```
    android:required="true" />
```

HW – Hint – Layout



```
<?xml version="1.0" encoding="utf-8"?>
<fragment xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/details_fragment"
    android:name="com.example.androidtvapp.VideoDetailsFragment"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".DetailsActivity"
    tools:deviceIds="tv" />
```

```
<?xml version="1.0" encoding="utf-8"?>
<fragment xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main_browse_fragment"
    android:name="com.example.androidtvapp.MainFragment"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"
    tools:deviceIds="tv"
    tools:ignore="MergeRootFrame" />
```

HW – Hint



```
DetailsActivity.java x DetailsDescriptionPresenter.java x MainActivity.java x CardPresenter.java
1 package com.example.androidtvapp;
2
3 import ...
4
5
6 /*
7  * Main Activity class that loads {@link MainFragment}.
8  */
9 public class MainActivity extends Activity {
10
11     @Override
12     public void onCreate(Bundle savedInstanceState) {
13         super.onCreate(savedInstanceState);
14         setContentView(R.layout.activity_main);
15     }
16 }
```

```
DetailsActivity.java x DetailsDescriptionPresenter.java x CardPresenter.java x activity_details
1 package com.example.androidtvapp;
2
3
4 import android.app.Activity;
5 import android.os.Bundle;
6
7 /*
8  * Details activity class that loads LeanbackDetailsFragment class
9  */
10 public class DetailsActivity extends Activity {
11
12     public static final String MOVIE = "Movie";
13
14
15     /**
16      * Called when the activity is first created.
17      */
18     @Override
19     public void onCreate(Bundle savedInstanceState) {
20         super.onCreate(savedInstanceState);
21         setContentView(R.layout.activity_details);
22     }
23
24 }
```

HW – Hint



```
DetailsActivity.java × DetailsDescriptionPresenter.java × CardPresenter.java × activity_details.xml ×
1
2
3 package com.example.androidtvapp;
4
5 import androidx.leanback.widget.AbstractDetailsDescriptionPresenter;
6
7 public class DetailsDescriptionPresenter extends AbstractDetailsDescriptionPresenter {
8     @Override
9     protected void onBindDescription(ViewHolder viewHolder, Object item) {
10         Movie movie = (Movie) item;
11
12         if (movie != null) {
13             viewHolder.getTitle().setText(movie.getTitle());
14             viewHolder.getSubtitle().setText(movie.getStudio());
15             viewHolder.getBody().setText(movie.getDescription());
16         }
17     }
18 }
19 }
```

HW – Movie



```
DetailsActivity.java × CardPresenter.java × Movie.java × activity_details.xml ×
package com.example.androidtvapp;

import java.io.Serializable;
import java.net.URI;
import java.net.URISyntaxException;

public class Movie implements Serializable {
    static final long serialVersionUID = 727566175075968653L;
    private long id;
    private String title;
    private String studio;
    private String description;
    private String cardImageUrl;

    public long getId() { return id; }

    public void setId(long id) { this.id = id; }

    public String getTitle() { return title; }

    public void setTitle(String title) { this.title = title; }

    public String getStudio() { return studio; }

    public void setStudio(String studio) { this.studio = studio; }

    public String getDescription() { return description; }

    public void setDescription(String description) { this.description = description; }

    public String getCardImageUrl() { return cardImageUrl; }

    public void setCardImageUrl(String cardImageUrl) { this.cardImageUrl = cardImageUrl; }

    public Movie() {
    }
}
```

```
public URI getCardImageURI() {
    try {
        return new URI(getCardImageUrl());
    } catch (URISyntaxException e) {
        return null;
    }
}

@Override
public String toString() {
    return "Movie{" +
        "id=" + id +
        ", title=" + title + '\'' +
        ", cardImageUrl=" + cardImageUrl + '\'' +
        '}';
}
```

HW – Hint - CardPresenter



```
public class CardPresenter extends Presenter {

    private static final String TAG = CardPresenter.class.getSimpleName();

    private static Context mContext;
    private static int CARD_WIDTH = 313;
    private static int CARD_HEIGHT = 176;

    static class ViewHolder extends Presenter.ViewHolder {
        private Movie mMovie;
        private ImageCardView mCardView;
        private Drawable mDefaultCardImage;
        private PicassoImageCardViewTarget mImageCardViewTarget;

        public ViewHolder(View view) {
            super(view);
            mCardView = (ImageCardView) view;
            mImageCardViewTarget = new PicassoImageCardViewTarget(mCardView);
            mDefaultCardImage = mContext.getResources().getDrawable(R.drawable.movie);
        }

        public void setMovie(Movie m) { mMovie = m; }

        public Movie getMovie() { return mMovie; }

        public ImageCardView getCardView() { return mCardView; }

        public Drawable getDefaultCardImage() { return mDefaultCardImage; }

        protected void updateCardViewImage(Uri uri) {
            Picasso.with(mContext).Picasso
                .load(uri.toString())
                .requestCreator()
                .resize(Utils.convertDpToPixel(mContext, CARD_WIDTH),
                    Utils.convertDpToPixel(mContext, CARD_HEIGHT))
                .placeholder(mDefaultCardImage)
                .error(mDefaultCardImage)
                .into(mImageCardViewTarget);
        }
    }
}
```

```
@Override
public ViewHolder onCreateViewHolder(ViewGroup parent) {
    Log.d(TAG, msg: "onCreateViewHolder");
    mContext = parent.getContext();

    ImageCardView cardView = new ImageCardView(mContext);
    cardView.setCardType(BaseCardView.CARD_TYPE_INFO_UNDER);
    cardView.setInfoVisibility(BaseCardView.CARD_REGION_VISIBLE_ALWAYS);
    cardView.setFocusable(true);
    cardView.setFocusableInTouchMode(true);
    cardView.setBackgroundColor(mContext.getResources().getColor(R.color.fastlane_background));
    return new ViewHolder(cardView);
}

@Override
public void onBindViewHolder(Presenter.ViewHolder viewHolder, Object item) {
    Movie movie = (Movie) item;
    ((ViewHolder) viewHolder).setMovie(movie);

    Log.d(TAG, msg: "onBindViewHolder");
    if (movie.getCardImageUrl() != null) {
        ((ViewHolder) viewHolder).mCardView.setTitleText(movie.getTitle());
        ((ViewHolder) viewHolder).mCardView.setContentText(movie.getStudio());
        ((ViewHolder) viewHolder).mCardView.setMainImageDimensions(CARD_WIDTH, CARD_HEIGHT);
        ((ViewHolder) viewHolder).updateCardViewImage(movie.getCardImageURI());
        //((ViewHolder) viewHolder).mCardView.setMainImage(((ViewHolder) viewHolder).getDefaultCardImage());
    }
}

@Override
public void onUnbindViewHolder(Presenter.ViewHolder viewHolder) {
    Log.d(TAG, msg: "onUnbindViewHolder");
}

@Override
public void onViewAttachedToWindow(Presenter.ViewHolder viewHolder) {
    // TO DO
}
```

HW – Hint



```
@Override
public void onViewAttachedToWindow(Presenter.ViewHolder viewHolder) {
    // TO DO
}

public static class PicassoImageCardViewTarget implements Target {
    private ImageCardView mImageCardView;

    public PicassoImageCardViewTarget(ImageCardView imageCardView) {
        mImageCardView = imageCardView;
    }

    @Override
    public void onBitmapLoaded(Bitmap bitmap, Picasso.LoadedFrom loadedFrom) {
        Drawable bitmapDrawable = new BitmapDrawable(mContext.getResources(), bitmap);
        mImageCardView.setMainImage(bitmapDrawable);
    }

    @Override
    public void onBitmapFailed(Drawable drawable) { mImageCardView.setMainImage(drawable); }

    @Override
    public void onPrepareLoad(Drawable drawable) {
        // Do nothing, default_background manager has its own transitions
    }
}
```




References

- <https://web.stanford.edu/>
- <https://developer.android.com/>
- <https://introcs.cs.princeton.edu/>
- <https://www.cs.utexas.edu/>
- <https://www.codepath.com/>
- <https://geeksforgeeks.com/>
- proandroidev



Thank you for listening

*"Coming together is a beginning;
Keeping together is progress;
Working together is success."*
- HENRY FORD